

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE	
	Regular End Semester Examination – Winter 2022	
	Course: B. Tech.	Branch : Mechanical
		Semester : V
	Subject Code & Name: BTMOE505D(Product Design Engineering)	
	Max Marks: 60	Date: 14/02/2023
		Duration: 3.00 Hr.
	Instructions to the Students:	
	1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.	
		(Level/CO)
		Marks
Q. 1	Solve Any Two of the following.	
A)	Define the product. Explain the various reasons that trigger new product development.	CO1
B)	Write a short note on reverse engineering .What is the benefits and limitation of reverse engineering approach to product design.	CO1
C)	Discuss in brief the problem solving approach for Product Design.	CO1
Q.2	Solve Any Two of the following.	
A)	Define ideation. Explain any two techniques of ideation.	CO2
B)	Define product architecture. Differentiate between the integral and modular architecture.	CO2
C)	What is Market research? Explain the importance of Market research in Product Design Engineering.	CO2
Q. 3	Solve Any Two of the following.	
A)	Define prototype. What are different types of prototypes?	CO3
B)	New Product Development is task of Multidisciplinary team. Explain the statement in detail with an example.	CO1
C)	Discuss in brief the health and hazard related issues of process operations in industries with suitable examples.	CO3
Q.4	Solve Any Two of the following.	
A)	What is geometrical modelling? Explain wireframe, solid and surface modelling with sketches.	CO4
B)	Write a short note on CAD/CAM.	CO4
C)	What is the purpose of materials and safety data-sheet? What is the	CO4

	minimum content of the materials and safety data-sheet?		
Q. 5	Solve Any Two of the following.		
A)	Explain the Product Life Cycle management approach related to environmental concerns of a product.	CO5	6
B)	Write a short note on sustainable waste management.	CO5	6
C)	Explain in brief the concepts of recycle and reuse with a case study.	CO5	6
	*** End ***		

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